IMU Kinematic

X_0 : initial IMU ~~reading~~ reading = [quaternion, angular velocity]^T.

X : state ~~derivat~~ deviation = $(\Delta X_{des} - \Delta X_0)$

ΔX_0 : state displacement after Initial reading.

ΔX_{des} : desired state displacement (for static, $\omega = 0$)

body level PD: $\alpha_{correction} = K_p \cdot$ (~~displace~~ angular displacement) $+ K_d$ (angular velocity).

$q = \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix}$, rotation about a unit axis with angle θ: $\vec{a} = (a_x, a_y, a_z)^T$

$$q = \begin{bmatrix} \cos(\frac{\theta}{2}) \\ a_x \sin(\frac{\theta}{2}) \\ a_y \sin(\frac{\theta}{2}) \\ a_z \sin(\frac{\theta}{2}) \end{bmatrix}$$

for rotating a 3D vector: $P = [P_x, P_y, P_z]^T$

represent it as: $p = \begin{bmatrix} 0 \\ P_x \\ P_y \\ P_z \end{bmatrix}$. then rotate by unit ~~quat~~ quaternion:

$$q = \begin{bmatrix} \cos(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) \\ \sin(\frac{\theta}{2}) \end{bmatrix} \rightarrow p_{(rotated)} = q \otimes p \otimes q^*$$

where $q^* = q^{-1}$ for unit quaternions.

$q = q_z(\psi)\, q_y(\theta)\, q_x(\phi)$

$q_z(\psi) = (\cos(\frac{\psi}{2}), 0, 0, \sin(\frac{\psi}{2}))$ $q_y(\theta) = (\cos(\frac{\theta}{2}), 0, \sin(\frac{\theta}{2}), 0)$.

$q_x(\phi) = (\cos(\frac{\phi}{2}), \sin(\frac{\phi}{2}), 0, 0)$

Initial state: (set as reference) $X_0 = \begin{bmatrix} q_0 \\ w_0 \end{bmatrix}$

current state: $X_{curr} = \begin{bmatrix} q \\ w \end{bmatrix}$.

Xdes (desire state) $= \begin{bmatrix} \Delta q_{des} \\ \Delta w_{des} \end{bmatrix}$.

state deviation:

angular displacement:

~~$(X_{current} - X_0)(q_{curr} - q_0)$~~

$\Delta q = q_0^{-1} \cdot q \rightarrow$ state deviation $= \Delta q_{des}^{-1} \cdot (\Delta q)$

angular velocity: $\Delta w = w - w_0 \rightarrow$ state dev $= (\Delta w_{des} - \Delta w)$.

~~ll~~ state ~~dus~~ dev: $\Delta X = \begin{bmatrix} \Delta q_{des}^{-1} \otimes (q_0^{-1} \otimes q) \\ \Delta w_{des} - (w - w_0) \end{bmatrix}$

~~body~~ RP: ~~$\alpha_{body} = K_p$~~

$\Delta Q = \Delta q_{des}^{-1} \otimes (q_0^{-1} \otimes q) = \left[\cos(\frac{\theta}{2}) \cdot \hat{a}\sin(\frac{\theta}{2})\right]$ $\hat{a}$ is unit vector.

define $\Delta Q = (w, v) \rightarrow w = \cos(\frac{\theta}{2})$, $v = \hat{a}\sin(\frac{\theta}{2})$.

$|V| = |\sin(\frac{\theta}{2})| = \sin(\frac{\theta}{2}) \rightarrow \frac{|V|}{w} = \tan(\frac{\theta}{2})$

$\theta = 2\,\mathrm{atan2}(|V|, w)$.

$\theta\hat{a} = 2\,\mathrm{atan2}(|V|, w) \cdot \frac{V}{|V|}$

$$\alpha_{body} = k_p \cdot (\theta \hat{a}) - k_d \cdot (\omega_{body})$$

↓ integrate

$$\Delta W_{body} = \alpha_{body} \cdot dt. \quad \left(W^{B}_{body_{corr_k}} = W^{B}_{body_{corr_{k-1}}} + \alpha^{B}_{corr_k} \cdot \Delta t \right.$$

↓

transport theorem: let r_i^B = foot position vector in leg frame.

$$\frac{d P_{body}}{dt} = \frac{d P_{leg}}{dt} + \omega_{body} \times P^B$$

$$V^B_{foot} = -\omega^B_{corr} \times r_i^B.$$

derivation: → world frame body pos

world frame foot pos: $P_F^W = P_B^W + R_{WB} r_F^B$ → body frame foot pos.

$$\frac{dP_F^W}{dt} = \frac{d P_B^W}{dt} + R_{WB}\left(\frac{d r_F^B}{dt} + \omega_B^B \times r_F^B\right)$$ transport theorem.

if static: $\frac{d P_B^W}{dt} = \frac{dP_F^B}{dt} = \frac{dP_F^W}{dt} = 0.$

then $\frac{dP_B^W}{dt} = -R_{WB}\left(\frac{dr_F^B}{dt} + \omega_B^B \times r_F^B\right)$

if no body translation: $\frac{dP_B^W}{dt} = 0.$

then $-R_{WB}\frac{dr_F^B}{dt} = R_{WB} \cdot (\omega_B^B \times r_F^B)$

$$\frac{dr_F^B}{dt} = -\omega_B^B \times r_F^B \rightarrow \boxed{\dot{r}_F^B = -\omega_B^B \times r_F^B}$$

then convert

$$\dot{r}_F^{\,B} = -\omega_B^{\,B} \times r_F^{\,B}$$

convert required leg velocity from body frame to leg frame:

since $r_F^{\,L} = r_F^{\,B} - \text{offset (translational)}$

$$\frac{dr_F^{\,L}}{dt} = \frac{dr_F^{\,B}}{dt} - 0 = \frac{dr_F^{\,B}}{dt}.$$

so $\dot{r}_F^{\,B} = \dot{r}_F^{\,L}$

then: joint correction velocity require:

$$\dot{q}_{joint} = J^{\#} \cdot \dot{r}_F^{\,L}$$

where $J^{\#} = J^T (JJ^T + \lambda^2 I)^{-1}$

then integrate: $q_{k+1} = q_k + \dot{q}_k \cdot \Delta t$.